Cyclic-voltage elimination of Pasch supports in two cubic configurations

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Abstract

We specialize the classical voltage and cycle-consistency method to two fixed cubic, linear 10_3 configurations. For an ordinary N-fold monomial cover, every elementary $(4,0)$ support projects injectively to a base Pasch and contributes exactly N supports precisely when its three fundamental cycle voltages vanish. Explicit unit normal forms over the integers give complete label histograms for every $N \geq 1$, including composite N. The numbers of eliminating raw label arrays are $N^{30} - N^{27}$ and $N^{30} - 5N^{27} + 10N^{24} - 10N^{21} + 4N^{19}$. We distinguish elimination in a cover sector from elimination in the full scalar-right CSS construction. The fixed seven-coordinate scalar forces at least seventy supports per Tanner graph. Replacing its length by $7q$ and changing just two base-edge labels yields a family with parameters $[[140q, 18, 6]]$, $q \geq 2$, column weight three, row weight six and no $(4,0)$ supports in either graph. The contribution is an explicit specialization and reproducible proof artifact; the general design method and the base configurations are classical. A bounded prior-art search does not establish priority for the exact specialization or code parameters.

Verification. Every manuscript proof has a checked Lean version. The 70 mathematical modules were built from their recorded sources; the axiom audit covers all 1,755 named authored declarations. The source links below are pinned to `41307a70c2c0`. The machine-readable proof inventory records the claim-to-theorem correspondence; the full file index links every compiled proof and preserved research import.

1 Question, prior art and conventions

The project-derived question is to characterize and count the edge labels that eliminate all elementary $(4,0)$ supports in the ordinary cyclic covers of two specific cubic configurations. It is a follow-up to an already completed incidence-corrected counting investigation, rather than a newly inventoried question posed by a source author. Abiad–Castriota [9] supplies the QC-GHP context and the earlier counting direction.

A variable neighborhood is a finite set $H(v) \subseteq C$. We assume $|H(v)| = 3$ and $|H(u) \cap H(v)| \leq 1$ for $u \neq v$. An elementary $(4,0)$ support is an *unordered* set of four variables for which every used check has selected degree two. In a cubic linear incidence structure it is a Pasch: the normal graph is K_4, with six distinct pair checks. Equivalently, its four actual binary columns have zero sum. A selected degree-four check would consume the unique common check of every pair and prevent pairing the remaining sockets, so a four-column binary dependency is automatically elementary under these hypotheses. We use the preserved generic Lean classification and actual binary-matrix bridge for these facts.

For CSS parameters we use the binary check-matrix model: $XZ^{\mathsf{T}} = 0$, $k = n - \operatorname{rank} X - \operatorname{rank} Z$, and distance equal to the minimum union support of a nontrivial logical pair (x, z). Such a pair obeys $Zx = Xz = 0$ and lies outside the product of the two stabilizer row spaces. The Lean statements use these actual matrices and this mixed-Pauli convention.

LDPC constructions based on voltage graphs were studied by Kelley and Walker [3]. Asvadi et al. [4] explicitly relate cyclic voltage sums to lifted cycle closure and intentional removal of trapping structures. The cycle-consistency matrix approach of Wang et al. [5, 6] gives the corresponding fundamental-cycle framework for absorbing structures. Its SCB separability, prime-field and consistency hypotheses must be compared with the present arbitrary monomial labels over $\mathbb{Z}/N\mathbb{Z}$; we prove that translation rather than treating the settings as identical. Earlier combinatorial QC constructions and Pasch-based distance arguments also appear in Vasić and Djordjević [1] and Noda [2]. For those two items and Kelley–Walker we inspected primary abstracts or metadata, not the unavailable full articles. They remain explicit limits on an exact novelty comparison.

The two base configurations below are classical; their construction and author data are credited to [7, 8]. Relevant passages and the author data were compared in the preceding research block, with exact relabelings retained in its artifact. We do not claim new configurations, a new anti-Pasch principle, a code-parameter record, decoder improvement or optimality.

The bounded search additionally covered public GitHub repositories and code, and three authenticated, read-only X Latest queries. All three X queries returned no results. GitHub searches had stated caps; selected public candidates led to the earlier 2003 and 2006 sources. Negative results from invalid or ineffective query qualifiers were excluded. Query strings, candidate inspections, source versions, caps and access failures are in the prior-art ledger. No comprehensive citation harvest or exhaustive code-equivalence search is claimed.

2 Projection and the exact voltage criterion

Let G be a finite abelian group and put a voltage $z(v,c) \in G$ on every existing incidence. Define the actual cover columns by

$$H_z(v,t) = \{(c, t + z(v,c)) : c \in H(v)\}. \tag{1}$$

Voltages on nonedges have no effect. The numerical counts below specialize to $G = \mathbb{Z}/N\mathbb{Z}$, $N \geq 1$.

Lemma 2.1 (Faithful Pasch projection)**.** *Every Pasch in H_z has four distinct base variable types and projects to a base Pasch.*

Proof. Two different fibers of the same variable cannot share a check: their single edges to a fixed base check would require $t + z = s + z$, hence $t = s$. The four pairwise adjacent variables of a lifted Pasch therefore have distinct types. Suppose three projected types contained the same base check c. Base linearity forces their pair checks to have type c. The single c-edge at one lifted variable then forces a common actual check of selected degree at least three, a contradiction. Thus projected selected degrees are at most two. Cubicity and pair completeness now give the base Pasch. □

Fix an ordered representative v_0, v_1, v_2, v_3 of a base Pasch, and let c_{ij} be its unique pair check. Set $d_{ij} = z(v_i, c_{ij}) - z(v_j, c_{ij})$. The six actual lifted pair incidences are the equations $x_j - x_i = d_{ij}$.

Theorem 2.2 (Voltage criterion and unordered count)**.** *The representative lifts if and only if*

$$d_{01} + d_{12} = d_{02}, \qquad d_{01} + d_{13} = d_{03}, \qquad d_{02} + d_{23} = d_{03}. \tag{2}$$

A balanced base support has exactly $|G|$ lifted supports. Consequently

$$\mathsf{N}_{4,0}(H_z) = |G| \, \#\{\textit{balanced base Pasch supports}\}. \tag{3}$$

Proof. Cancellation gives necessity. Under (2), all six equations hold for, and only for, $(x_0, x_1, x_2, x_3) = (t, t + d_{01}, t + d_{02}, t + d_{03})$, $t \in G$. The base pair checks are distinct, so these equations give an actual Pasch. Distinct t give distinct unordered supports because their base types differ; different base

supports have different projections. Lemma 2.1 gives surjectivity. Formally, the last step is a bijection between unordered supports with injective projection and a base support together with offsets above a fixed labeling. Reindexing that labeling changes neither existence nor multiplicity. The checked proof includes this finite-set partition, not just the solution count for a fixed ordered tuple. □

3 Complete label counts for the two fixed bases

Write a triple such as 012 for $\{0,1,2\}$. The star base has columns

$$012, 034, 135, 245, 067, 168, 269, 378, 479, 589.$$

The triangle base replaces its first four triples by $345, 125, 024, 013$. Both have ten variables, ten checks and thirty edges, and are cubic and linear on both sides. The star has the sole Pasch 0123. The triangle has exactly

$$0123, \quad 0789, \quad 1569, \quad 2468, \quad 3457.$$

These complete finite lists and all pair checks are kernel-checked. A raw label array has thirty coordinates, ordered by variable and then check. For the auxiliary eleven-coordinate slice, the allowed edge positions are $7, 10, 11, 16, 19, 22, 23, 25, 26, 28, 29$ for the star and $8, 10, 11, 16, 19, 22, 23, 25, 26, 28, 29$ for the triangle (zero-based indices); the remaining nineteen labels are zero. The label encodings are injective. We count raw arrays and these explicit slices independently. No assertion about equivalence classes under switching is needed or made.

For a subset J of the balance events, stack its three integer cycle rows into A_J. Explicit elementary integer row and column operations reduce A_J, both on all thirty edges and on the eleven slice coordinates, to $\mathrm{diag}(I_r, 0)$. For the star $r = 0, 3$. For the triangle,

$\lvert J\rvert$	0	1	2	3	4	5
r_J	0	3	6	9	11	11

The nonzero invariant factors are all units. This integer assertion is stronger than a rational or prime-field rank computation.

Lemma 3.1 (Certificate soundness)**.** *If integer matrices U, U^{-1}, V, V^{-1} satisfy the recorded inverse identities and $UAV = \mathrm{diag}(I_r, 0)$, reduction modulo every positive N gives N^{n-r} solutions to $Az = 0$ in $(\mathbb{Z}/N\mathbb{Z})^n$.*

Proof. The integer identities survive the ring homomorphism to $\mathbb{Z}/N\mathbb{Z}$. The map $z \mapsto V^{-1}z$ is a bijection from the original kernel to the diagonal kernel, with inverse $y \mapsto Vy$. The diagonal kernel consists of vectors with its first r coordinates zero and its remaining $n - r$ arbitrary. This argument needs neither a field nor coprimality. All inverse and normal-form identities, as well as their connection to the literal cycle rows, are checked in Lean. An independent Python verifier reconstructs and replays the same certificates. □

Theorem 3.2 (All-N label histograms)**.** *The raw numbers eliminating every left-cover Pasch are*

$$E_{\mathrm{star}}(N) = N^{30} - N^{27}, \tag{4}$$

$$E_{\mathrm{triangle}}(N) = N^{30} - 5N^{27} + 10N^{24} - 10N^{21} + 4N^{19}. \tag{5}$$

For the triangle, let g_j count slice arrays with exactly j balanced base Pasches. Then

$$g_0 = N^{11} - 5N^8 + 10N^5 - 10N^2 + 4,$$
$$g_1 = 5(N^8 - 4N^5 + 6N^2 - 3),$$
$$g_2 = 10(N^5 - 3N^2 + 2),$$
$$g_3 = 10(N^2 - 1), \qquad g_4 = 0, \qquad g_5 = 1.$$

Raw histograms are $N^{19}g_j$. The star slice histogram is $(N^{11}-N^8, N^8)$. An array in bin j gives Nj actual lifted supports.

Proof. Lemma 3.1 gives N^{n-r_J} for every event intersection. Inclusion-exclusion gives (4)–(5). More generally,

$$g_j = \sum_{s=j}^{5} (-1)^{s-j} \binom{s}{j} \binom{5}{s} N^{11-r_s}.$$

This follows pointwise: a truth pattern with k events contributes $\sum_{s=j}^{k} (-1)^{s-j} \binom{s}{j}\binom{k}{s}$, which is one for $k = j$ and zero otherwise. Summing gives the histogram. The Lean proof checks the finite truth-pattern identity, proves the summation bridge and groups subsets by cardinality. It then evaluates the six polynomials. Literal voltage data connect the abstract linear forms to the actual edge labels of both bases. The raw calculation uses $n = 30$, the slice calculation $n = 11$; their ratio and the displayed raw factors follow from the two independently checked counts. □

At $N = 1$ no base Pasch can be eliminated. For $N \geq 2$ the eliminating counts are positive. In particular,

$$g_0 = N^8(N^3-5) + 10N^2(N^3-1) + 4 > 0.$$

For the star, $N^{30} - N^{27} = N^{27}(N^3-1) > 0$.

Proposition 3.3 (Simultaneous original and dual elimination)**.** *For the star base, the number of raw arrays eliminating both its original and its dual-cover Pasch is*

$$N^{30} - 2N^{27} + N^{24}. \tag{6}$$

Proof. The dual has its sole Pasch on check types $6, 7, 8, 9$. Pulling its voltages back to the original edges with a minus sign gives a second three-row event. The stacked six-row matrix has an explicit unit normal form of rank six. Each individual event has rank three. Lemma 3.1 and two-event inclusion-exclusion give (6). □

4 Full scalar-right incidence and the quantum family

Let $R_N = \mathbb{F}_2[x]/(x^N+1)$, and let $A_{cv} = x^{z(v,c)}$ on each base edge. The adjoint $*$ transposes a matrix and replaces x by x^{-1}. For $b = 1 + x + x^3$, use

$$X = [A \mid b^* I_C], \qquad Z = [bI_V \mid A^*]. \tag{7}$$

Thus $XZ^* = Ab^* + b^*A = 0$. Equivalently, common cyclic translations commute with the arbitrary monomial cover, and the two equal binary summands cancel. The Lean proof uses the actual expanded sparse matrices and block product.

An X left column (v,t) is (1); an X right column (c,t) has checks $(c, t-e)$ for $e \in \{0,1,3\}$. A Z left column (v,t) has checks $(v, t+e)$, and a Z right column (c,t) has checks $(v, t-z(v,c))$. These signs specify the ordinary binary transpose convention.

Theorem 4.1 (Full support partition)**.** *For $N \geq 7$, any base whose incidence and transpose are cubic and linear, and any monomial labels, suppose the scalar columns B are cubic and linear. Then*

$$\mathsf{N}_{4,0}(X) = \mathsf{N}_{4,0}(H_z) + |C|\mathsf{N}_{4,0}(B), \qquad \mathsf{N}_{4,0}(Z) = \mathsf{N}_{4,0}(H_z^*) + |V|\mathsf{N}_{4,0}(B). \tag{8}$$

No mixed-sector Pasch occurs. For the two ten-type bases and $N = 7$, each full Tanner graph therefore retains at least seventy supports, for every left label assignment.

Proof. An adjacent mixed pair has no proper common neighbor: base linearity and the single edge of each left variable at a fixed check type would force any such neighbor into a triple common check. In a Pasch every third variable would be a proper common neighbor of either root pair, so a mixed pair is impossible. Every support is all left or all right. Pairwise adjacency forces an all-right support into a single check-type fiber. The finite-set partition gives the sum of the left-cover count and one scalar count per fiber. The scalar at $N = 7$ has exactly seven supports; its transpose has the same count. Hence $10 \cdot 7 = 70$ is unavoidable even when every left-cover Pasch is eliminated. □

Theorem 4.2 (Explicit family)**.** *For every integer $q \geq 2$, put $N = 7q$, use the star base and set*

$$z(0,0) = z(4,6) = 7, \qquad z(v,c) = 0 \text{ on all other edges.} \tag{9}$$

The actual binary checks (7) define a CSS code with parameters $[[140q, 18, 6]]$. Both graphs have column weight three, row weight six, no Tanner four-cycle and no elementary $(4,0)$ support.

Proof. Scalar factor. The kernel equation of the negative-shift scalar is $f_{t+3} = f_t + f_{t+1}$. Every solution is determined by three bits (a, b, c) and follows the seven-periodic pattern

$$(a, b, c, a+b, b+c, a+b+c, a+c).$$

At length $7q$ all eight patterns occur. Every nonzero pattern has four ones per seven coordinates, so every nonzero scalar kernel word has weight $4q$. Thus $q \geq 2$ excludes weight four. Reflection proves the same assertion for the transposed scalar. The differences of $\{0, 1, 3\}$ are distinct modulo $N \geq 14$, proving scalar cubicity and linearity.

Cover sectors. The original sole Pasch has a cycle residual 7, and the dual sole Pasch has a residual ± 7; these are nonzero modulo $7q$. The checked duality maps base variable u to dual variable $u + 6 \pmod{10}$ and base check c to dual check $c + 4 \pmod{10}$, changing the two-edge label pattern to its negative. Hence both cover sectors have no four-column binary dependency. Theorem 4.1 excludes supports in the full graphs. Cubicity follows from the two three-socket column types. Cover linearity, scalar linearity and the one-edge-per-check-type property exclude Tanner four-cycles, including mixed pairs. A row has three cover neighbors and three scalar neighbors, giving weight six.

Dimension. In either check-equation kernel, the scalar conditions force each coordinate sequence into a seven-periodic three-bit pattern (with reflection for the negative orientation). All nonzero labels are multiples of seven, so shifts by those labels act trivially on such sequences. Each of the three phases therefore belongs to the kernel of the binary star incidence or its transpose. Each base kernel has exactly eight elements, as certified by its complete finite support list. Consequently each expanded check-equation kernel has $8^3 = 512 = 2^9$ elements. Binary rank-nullity gives

$$\operatorname{rank} X = \operatorname{rank} Z = 10N - 9, \qquad k = 20N - 2(10N - 9) = 18.$$

This recurrence argument also covers even q, without a square-free or coprimality assumption on the polynomial quotient.

Distance. Put ones on left variable types $\{2, 3, 5, 6, 7, 8\}$ at fiber zero and zero elsewhere, obtaining w of weight six. These six base columns sum to zero and avoid the two modified variables, so $Xw = 0$. Let p be the scalar pattern with initial bits $(1, 0, 0)$, and put $u(v, t) = p(-t)$ for left type $v = 2$, with zero on all other coordinates. Then $Zu = 0$ and $\langle w, u \rangle = 1$. Every vector in the row space of Z is orthogonal to u, so w is a nontrivial logical word. Column weight three forces every kernel word to have even weight; linearity excludes weight two; the elimination proof excludes weight four. Thus both classical check kernels have minimum nonzero weight at least six. Every nontrivial mixed CSS logical has a nontrivial component whose support is contained in its Pauli support, so its weight is at least six. The pure word w attains six, proving the exact distance. □

The family has constant dimension and distance; it is a structural example, not a claim of an asymptotically good or experimentally improved code. The case $q = 2$ is $[[280, 18, 6]]$. An independent C++ verifier reconstructs all columns, checks both ranks 131, commutation, degrees and linearity, and exhaustively tests all 250,654,530 four-subsets in each graph. It also checks the logical support with zero-based indices $28, 42, 70, 84, 98, 112$ by an augmented row-rank calculation. Python exact consumers additionally cover $q = 3, 4$. These computations are independent checks of the specified finite instances; the all-q conclusion uses the proof above.

5 Proof artifacts and reproducibility

The proof inventory below names checked entry points. All seven manuscript proof blocks, plus their finite data, source translations and auxiliary claims, have Lean versions with matching hypotheses and conclusions. The axiom audit queries every named authored declaration, including definitions containing proofs. Source files and all imported research modules are included in the repository. Compiled objects, third-party PDFs and local caches are not redistributed.

Claim or bridge	Direct repository proof file
Projection and actual incidence	`PaschLiftProjection.lean`; `PaschLiftCriterion.lean`
Unordered all-N count	`SupportVoltageCount.lean`; `CountRepresentatives.lean`
Complete base lists and physical voltages	`StarVoltageData.lean`; `TriangleVoltageData.lean`
Integer certificate soundness	`NormalFormCount.lean`; individual normal forms and row connections in the manifest
Histograms and raw label counts	`LabelHistograms.lean`; `ConcreteElimination.lean`
Label encodings and positivity	`LabelEncoding.lean`; `PositiveLabelCounts.lean`
Simultaneous star/dual count	`JointElimination.lean`
Full scalar-right count	`QCSupportCount.lean`; `FullGraphCounts.lean`
Fixed seven-coordinate obstruction	`FanoObstruction.lean`
Scalar factor, all q	`ScalarWeight.lean`; `ScalarIncidence.lean`
Actual CSS matrices and dimensions	`SparseCSSMatrices.lean`; `FamilyRank.lean`
Full-graph elimination and distance	`FamilyGraphs.lean`; `FamilyPauliDistance.lean`
Row degrees and final family	`SparseRowDegrees.lean`; `FamilyTheorem.lean`

The pinned environment is Lean `v4.36.0-rc1`, mathlib revision `5f7b4198d22a` (full revision in the manifest). The final modules were built from their recorded sources and audited for axioms; only `propext`, `Classical.choice` and `Quot.sound` are permitted. No admitted proof, native-decision axiom or custom mathematical axiom is accepted. The new checks distinguish newly compiled modules from the unchanged, previously freshly checked sixty-one-module dependency artifact.

For example, the following are ASCII renderings of checked entry-point statements. The first has $N > 0$; the last two have $q > 0$, and $F = \mathbb{Z}/2\mathbb{Z}$. Here `Logical` uses the two actual binary syndromes and excludes the corresponding stabilizer row spaces; `pauliWeight` counts the union of the two component supports.

```
theorem star_raw_count :
  (Nat.card {z : Fin 30 -> ZMod N //
    Eliminates StarVoltageData.H (StarVoltageData.labelsRaw z)}
    : Int) = (N : Int)^30 - (N : Int)^27
```

```
theorem dimension :
  Fintype.card (Qubits q) - (X q).rank - (Z q).rank = 18 := by
  have hx := rankX q
  have hz := rankZ q
  have hq := NeZero.ne q
  rw [length]
  omega

theorem distance_six (hq : 2 <= q) :
  And (forall x z : Qubits q -> F,
         Logical q x z -> 6 <= pauliWeight q x z)
      (And (Logical q 0 (word q))
           (pauliWeight q 0 (word q) = 6))
```

The source proofs respectively transport the exact physical elimination predicate to its counted balance event, apply binary rank-nullity to the literal check matrices, and combine a lower bound for every mixed Pauli word with the explicit separating logical witness. The complete statements, including their section parameters and imports, are in the linked files.

Run `verify.py` for the independent certificate and C++ checks, and `verify_lean.py --project /path/to/pinned/project` after rebuilding the preserved dependency artifact. The independent slice enumeration covers all eleven-coordinate arrays for $N = 2, 3, 4$ in both bases: 8,746,998 arrays. The polynomial histogram agrees in every case. The 128 seeded direct lifted incidence consumers are sampled checks, not exhaustive evidence. Frozen predecessor archives remain unchanged.

A Reference chronology and timing

Every delta is the age of the stated source milestone at this paper's revision, 24 October 2026. It is *not* a claim that a problem remained unsolved for that interval. Year-only dates give a day-range; conference intervals are retained. For the 2016 source we use the earliest located publisher date, not an asserted earliest preprint. Date arithmetic is reproduced in `reference-chronology.json`.

Source	Earliest located milestone	Inspected version / access	Delta (days)
Vasić–Djordjević	1–3 Oct. 2003	Conference interval; primary institutional abstract only	8422–8424
Noda	3–7 Apr. 2006	Conference interval; primary publisher abstract only	7505–7509
Kelley–Walker	2008 (year only)	ISIT year; institutional abstract only	6506–6871
Asvadi et al.	23 Feb. 2010	arXiv v1 read; final 2011 metadata only	6087
Wang et al.	2011 (year only)	ITA author PDF read	5411–5775
Wang et al.	30 Aug. 2012	5 Dec. 2012 v2 read; final 2013 metadata only	5168
Prażmowska and Prażmowski	16 Feb. 2016	Publisher date; relevant HTML proofs read	3903
Abreu et al.	10 Apr. 2021	12 Sep. 2021 v2 passages and author data	2023
Abiad–Castriota	15 July 2026	v1 source conventions rechecked	101

The bounded mathematical research block ran from 15:54:07 to 16:54:07 UTC on 24 October 2026 (sixty minutes). It included the prior-art audit and ordinary construction/count arguments. Subsequent

Lean proof transfer, manuscript preparation and publication are recorded separately; that later work is not presented as part of the sixty-minute investigation. Final publication timing and commit identifiers are recorded in the artifact's publication manifest.

B Scope of the answer

The general voltage-elimination method was already available in classical literature. The answered project-derived specialization consists of the exact fixed-base criteria and histograms, the full-incidence scalar obstruction and the explicit constant-dimension, constant-distance CSS family. These are not additional source-posed inventory entries and do not alter the canonical coverage denominator. The precise novelty comparison remains bounded by the recorded access and search gaps. Decoder behavior, error-floor performance, optimality and a comparison against controlled code families are outside this paper's conclusions.

References

[1] B. Vasić and I. B. Djordjević, Quasicyclic low-density parity check codes, TELSIKS 2003, vol. 2, pp. 417–420. doi:10.1109/TELSKS.2003.1246258. Primary institutional abstract and metadata inspected.

[2] M. Noda, Designing a Self-orthogonal Quasi-cyclic Code with Extended Minimum Hamming Distance, TURBO-CODING 2006, Munich, 3–7 April 2006. Publisher record. Abstract and metadata inspected; full six-page article unavailable.

[3] C. A. Kelley and J. L. Walker, LDPC codes from voltage graphs, ISIT 2008. doi:10.1109/ISIT.2008.4595095. Institutional record; abstract only.

[4] R. Asvadi, A. H. Banihashemi and M. Ahmadian-Attari, Lowering the Error Floor of LDPC Codes Using Cyclic Liftings, arXiv:1002.4311v1, 23 February 2010. Relevant preprint statements and proofs inspected; 2011 journal revision not compared in full.

[5] J. Wang, L. Dolecek and R. Wesel, LDPC Absorbing Sets, the Null Space of the Cycle Consistency Matrix, and Tanner's Constructions, ITA 2011. Author conference PDF.

[6] J. Wang, L. Dolecek and R. Wesel, The Cycle Consistency Matrix Approach to Absorbing Sets in Separable Circulant-Based LDPC Codes, arXiv:1208.6094v2, 5 December 2012; first preprint 30 August 2012. IEEE Trans. Inform. Theory 59(4) (2013), 2293–2314; final journal revision not compared in full.

[7] M. Prażmowska and K. Prażmowski, Binomial partial Steiner triple systems containing complete graphs, Graphs and Combinatorics 32 (2016), 2079–2092. doi:10.1007/s00373-016-1681-3. Relevant publisher-HTML definitions and proofs inspected in the preceding block.

[8] M. Abreu, M. Funk, V. Krčadinac and D. Labbate, Strongly regular configurations, arXiv:2104.04880v2, 12 September 2021; first preprint 10 April 2021. Designs, Codes and Cryptography 90 (2022), 1881–1897. Relevant preprint passages and author data inspected.

[9] A. Abiad and N. Castriota, Spectral and Additive Combinatorial Methods for Cycles and Absorbing Sets in Lifted-Product Quantum LDPC Codes, arXiv:2607.13666v1, 15 July 2026. Definition 3.1 and Section 3.2 matrix conventions rechecked; original counting context.